Network Administration

Lecture 3 - Linux Commands and Autonomous AI Agent

Joelle kasabreh | Student ID: 12428112 | Instructor: Jehad Hamamreh

An-Najah National University | First semester 2026-2027 | Ubuntu on Oracle VirtualBox

1. Introduction

For this assignment, I practiced using the Linux commands mentioned in Lecture 3 on an Ubuntu virtual machine and recorded my observations. I additionally tested a local AI assistant to autonomously select and execute limited system-check commands. My practice files were created in ~/Lecture3_Lab.

2. Linux Commands - File and Directory Practice

I used pwd, ls, ls -l, ls -a, mkdir and cd to navigate and view the contents of directories; cp, cp -r and mv to copy or move files; and rm, rmdir and rm -r to remove my temporary practice items. I also used cat, more and less to view text.

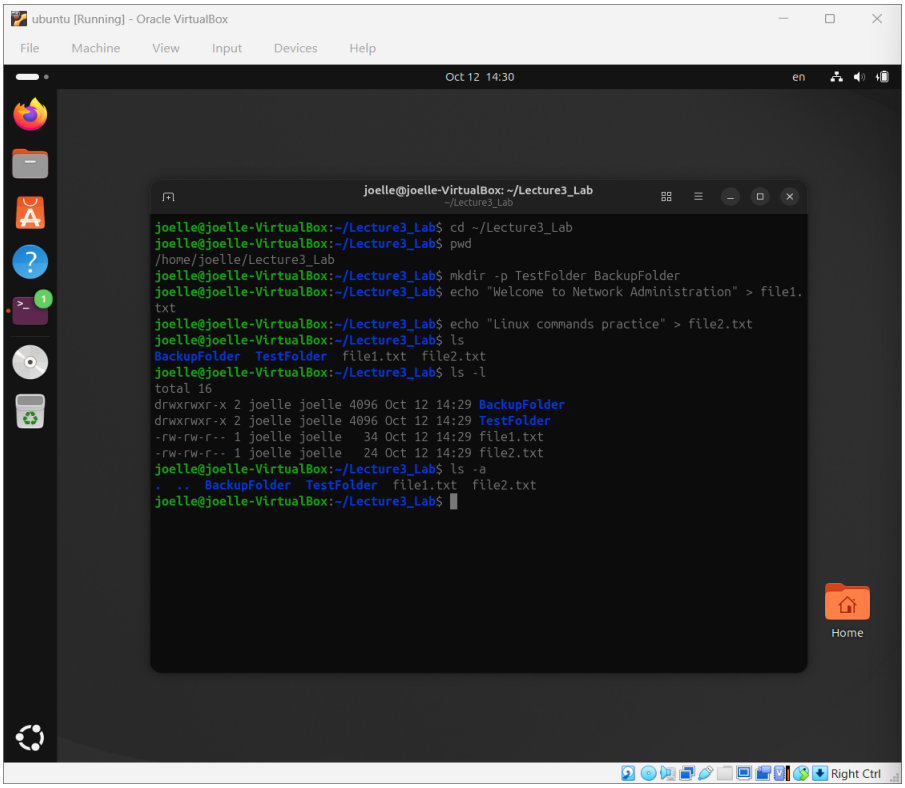


Figure 1. Creating and listing practice files.

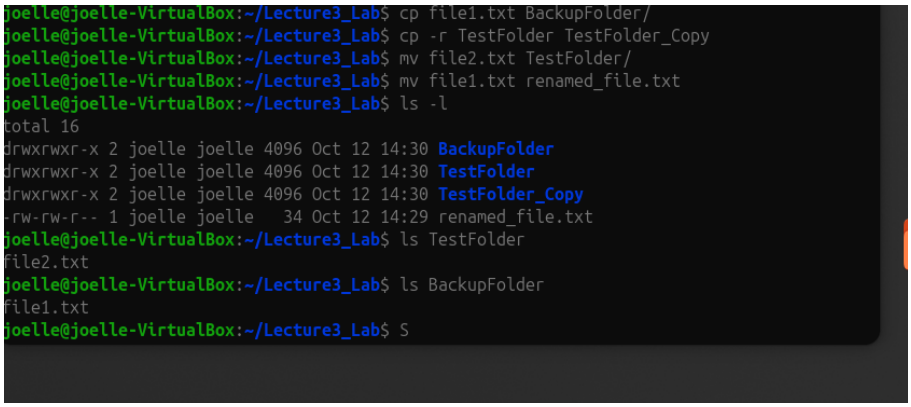


Figure 2. Copying, moving and renaming files.

3. Text Filtering and Processing

I prepared names.txt, students.txt and subjects.txt, then tested head, tail, wc, sort, sort -r, sort -t: -n -k 3, uniq, cut, grep, awk, paste and tr. Some of these commands helped me to count the number of lines, sort marks in ascending or descending order, find corresponding records and extract specific fields.

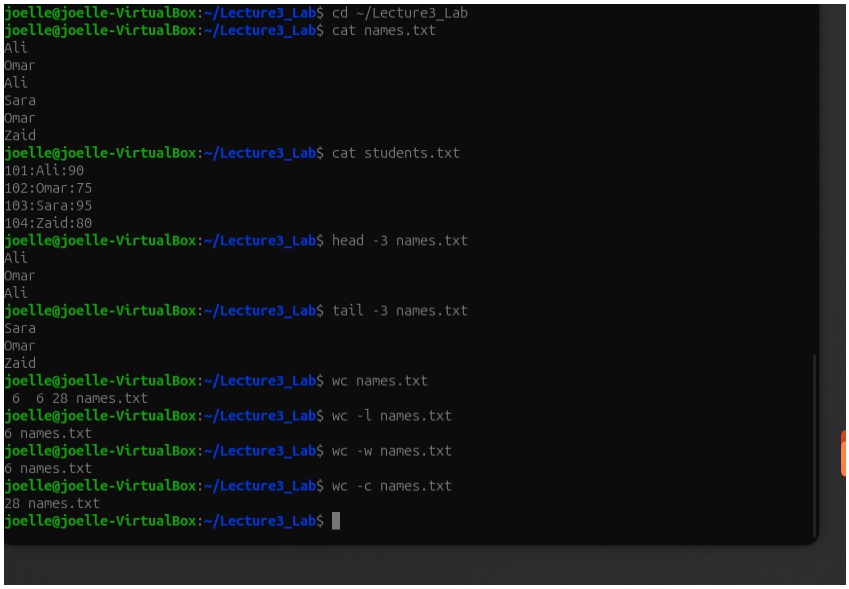


Figure 3. head, tail and wc results.

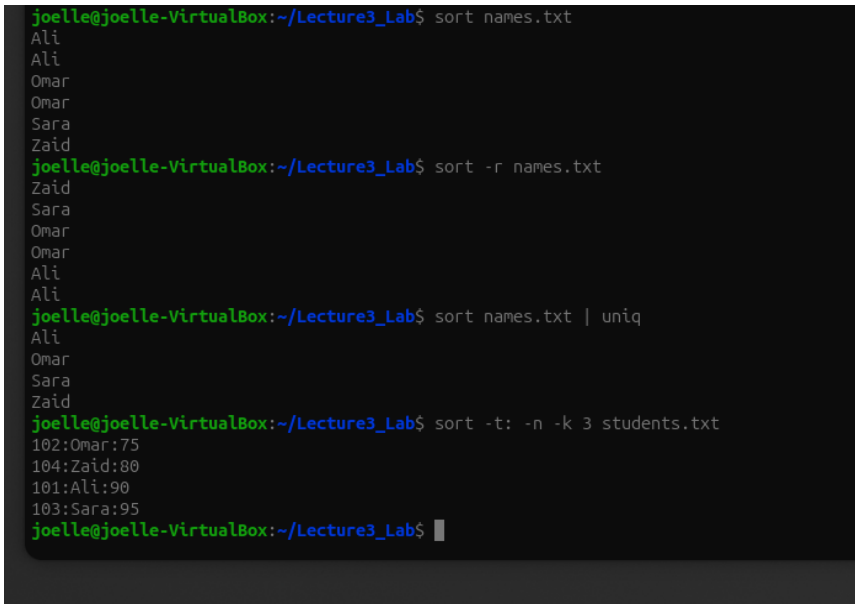


Figure 4. Alphabetical and numeric sorting.

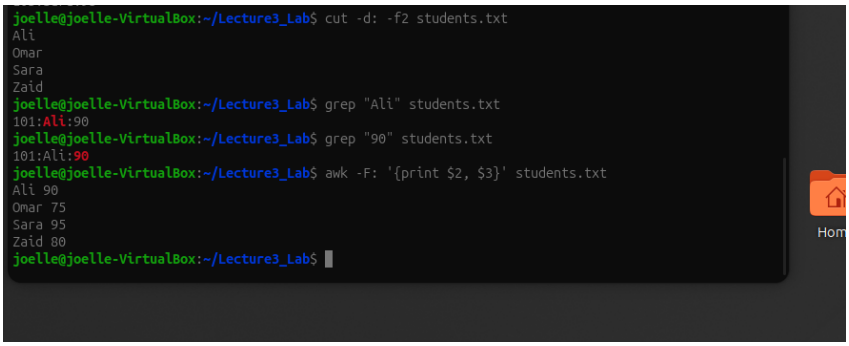


Figure 5. Extracting student data with cut, grep and awk.

The test of paste resulted in two entries with no subject because the source files have a different number of lines; this is an expected behavior.

4. Help, General Commands, Sessions and vi

I used man ls, man -k copy and man -K password for the documentation of commands; date, cal and echo for general purposes; and whoami, who and id to view account information. I reviewed the help screens of passwd and login instead of actually changing my password or initiating a full login session. bash --login, logout and exit were demonstrated in shell sessions.

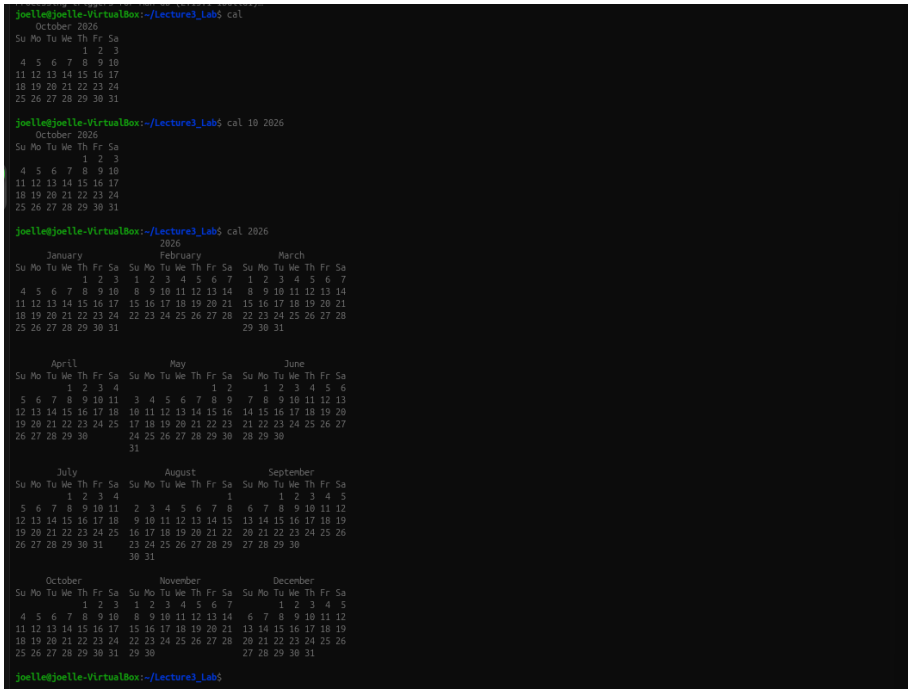


Figure 6. Displaying the monthly and yearly calendar.

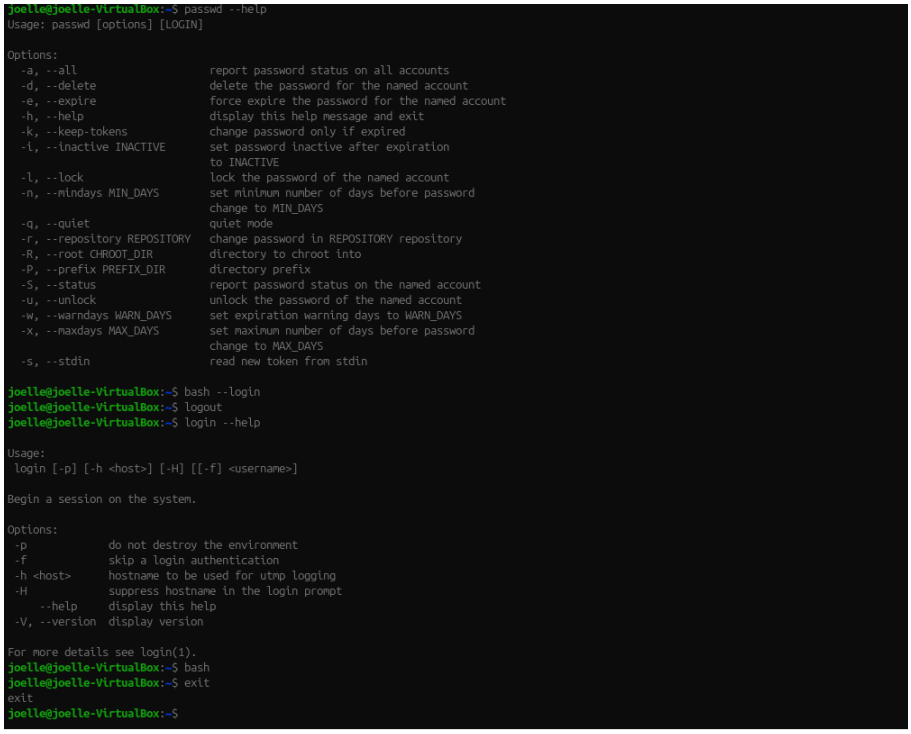


Figure 7. Account/session commands and help information.

With the vi editor, I created and edited the network_report.txt file, saved the changes and viewed its content with cat. The last editing attempt left in an extra character, which is evident in the output.

```
joelle@joelle-VirtualBox:~/Lecture3_Lab$ vi network_report.txt
joelle@joelle-VirtualBox:~/Lecture3_Lab$ cat network_report.txt
Network Administration - Lecture 3
Student: joelle kasabreh
Operating System: Ubuntu Linux
Virtualization: Oracle VirtualBox
Linux CLI Commands Practice
joelle@joelle-VirtualBox:~/Lecture3_Lab$ vi network_report.txt
joelle@joelle-VirtualBox:~/Lecture3_Lab$ cat network_report.txt
Network Administration - Lecture 3
gTask Status: Completed Successfully
Student: joelle kasabreh
Operating System: Ubuntu Linux
Virtualization: Oracle VirtualBox
Linux CLI Commands Practice
joelle@joelle-VirtualBox:~/Lecture3_Lab$
```

Figure 8. Saved text inspected after editing in vi.

5. AI Agent - Setup and Design

I installed the Ollama application and downloaded the qwen2.5:1.5b model (986 MB). Because the virtual disk initially had almost no free space, I expanded it from approximately 15 GB to 30 GB and added more space to the Ubuntu partition and the ext4 filesystem before downloading the model.

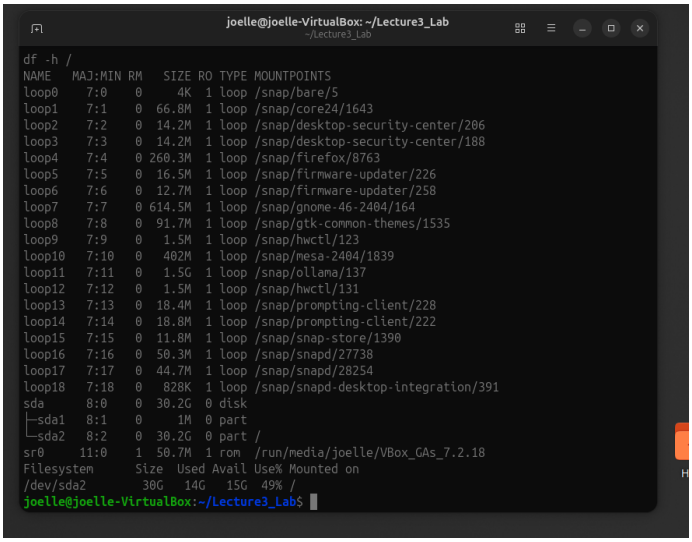


Figure 9. Virtual disk expanded; Ubuntu has more free space.

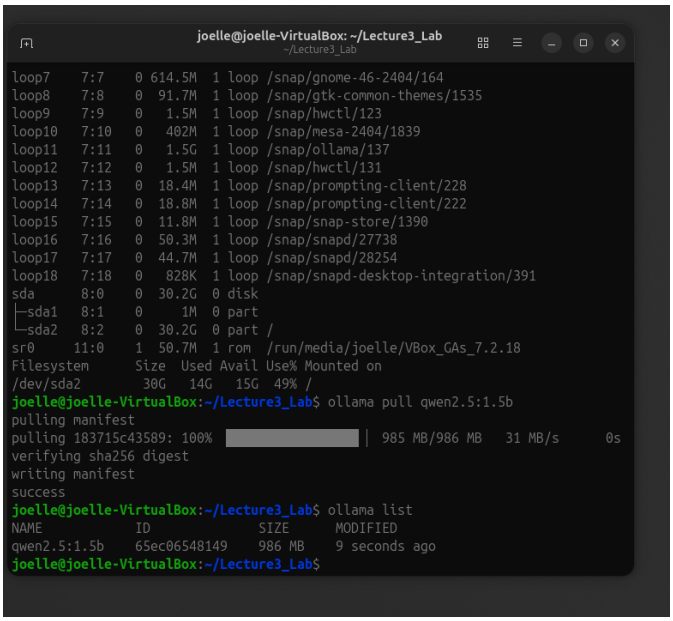


Figure 10. Ollama model downloaded and listed successfully.

How the agent works

My Python script, safe_ai_agent.py, requests the local model to select three checks from an approved list. It then executes the corresponding read-only commands, captures the actual terminal outputs, asks the model for an interpretation and saves the results in ai_health_report.txt. The model cannot execute arbitrary shell commands with this script.

Allowed checks: df -h /, free -h, uptime, uname -r, whoami and ip -brief address.

6. Autonomous Execution and Results

In the documented run, the AI agent picked disk, memory and network. It automatically executed df -h /, free -h and ip -brief address. The disk utilization was 52%, and the network interface enp0s3 was UP. The script then generated an analysis and saved the report.



Figure 11. AI-selected commands and their real output.

7. My Experience and Conclusion

The agent successfully selected and executed commands, and I didn't have to type every diagnostic command individually. However, the final interpretation of the agent was not fully reliable. It flagged the normal /dev/sda2 root partition as suspicious, questioned the swap size without adequate evidence and suggested an IPv6 issue that the output did not actually show. With this experience, I have learned that while AI can accelerate the process, an administrator should always verify the suggestions of an AI before making any changes to the system.

Overall, I successfully completed my command practice and demonstrated a working local AI agent. The main challenges were the limited virtual-machine resources and verifying the recommendations of the AI.

Source: Lecture 3 - Introduction to System Administration; screenshots from my Ubuntu terminal session.